



COM410 Programming in Practice

T8. Exception Handling



Exceptions

What is an exception?

- In Java, an exception is an unwanted or unexpected event, which occurs during the execution of a program that disrupts the normal flow of the program's instructions.
- Can be caught and handled by the program code
- Creates an Exception object that contains information such as the name and description of the exception

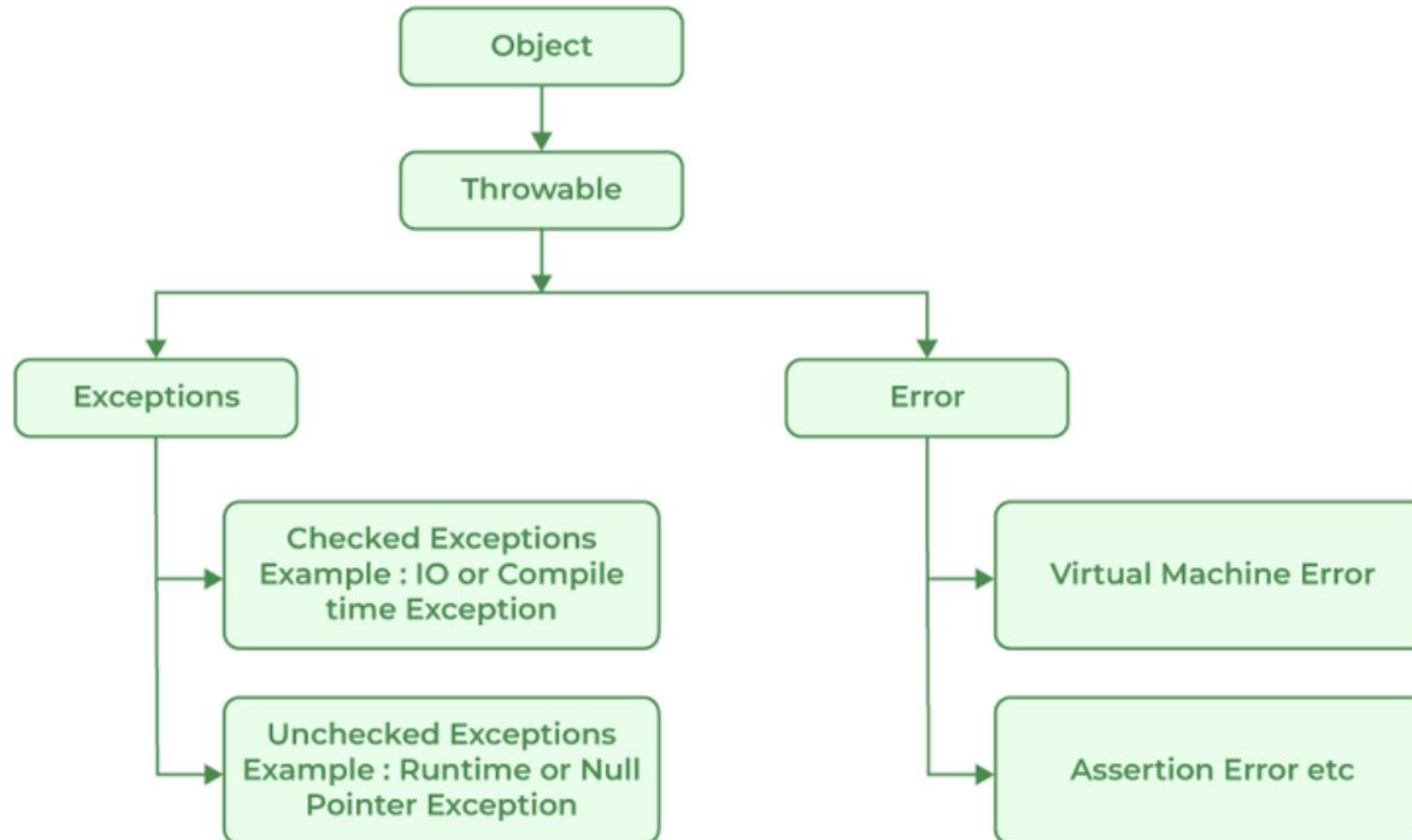
Exception vs Error

- **Exception:** a condition that a reasonable application might try to catch
- **Error:** a serious problem beyond the control of the programmer that a reasonable application should not try to catch

What causes exceptions?

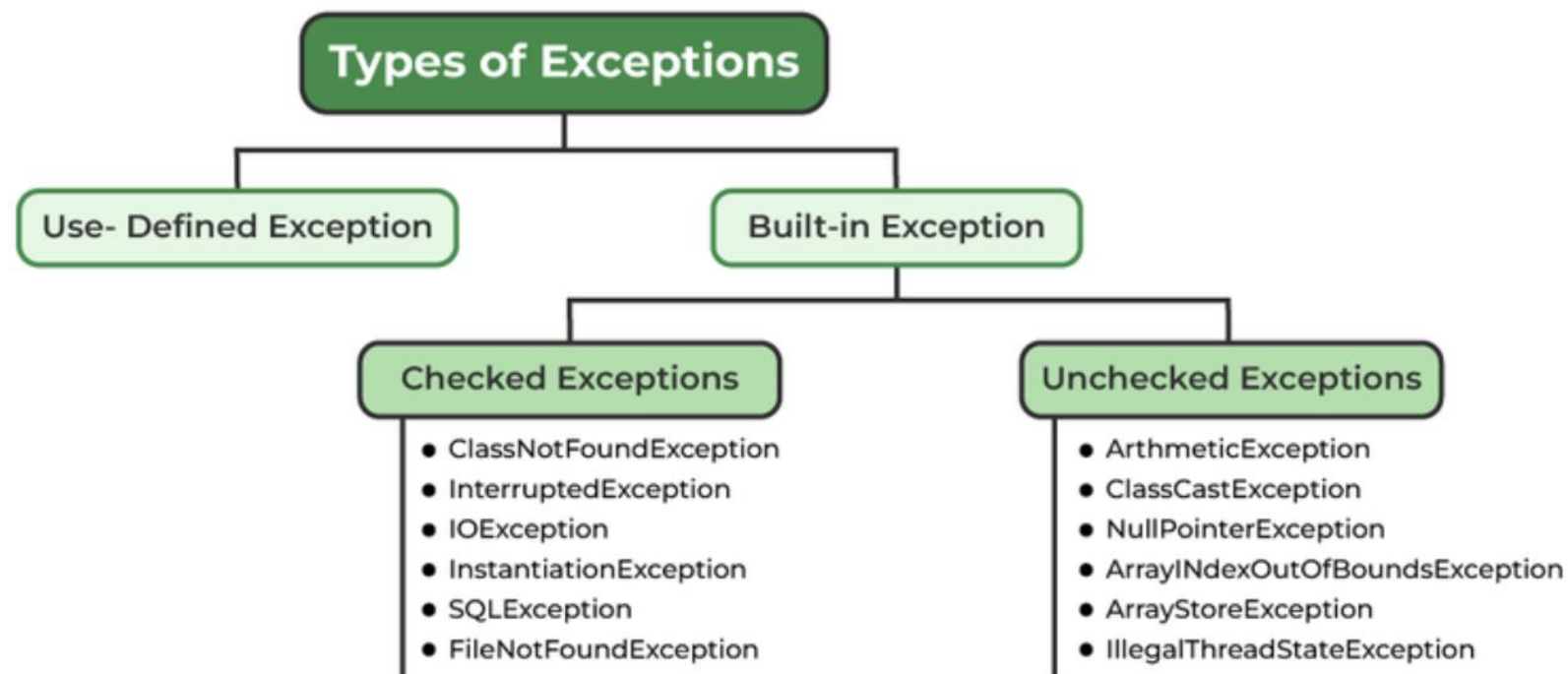
- Many reasons, including...
 - Invalid user input
 - Device failure
 - Loss of network connection
 - Opening an unavailable file
 - Code errors

Java Exception Hierarchy



Checked and unchecked exceptions

- Checked exceptions happen at compile time. Must be handled in the code or a compilation error is generated
- Unchecked exceptions do not generate compilation errors



Exception Handling

- File handling throws a checked exception which must be handled

```
import java.util.Scanner;
import java.io.File;

public class Demo {

    public static void main(String[] args) throws Exception {

        File file = new File( pathname: "src/textfile.txt");
        Scanner fileInput = new Scanner(file);

        while (fileInput.hasNextLine()) {
            System.out.println("Reading... " + fileInput.nextLine());
        }
    }
}
```

Exception Handling

- The `try...catch` block

```
try {  
    // some activity that might raise an exception  
} catch{  
    // code to deal with the exception  
}
```